Impact of Missing Data Mechanisms and Imputation Strategies on Machine-Learning-Based ICU Mortality Prediction

1. Introduction

Missing values are common in electronic health records (EHRs), particularly in intensive-care-unit (ICU) data, where measurements are collected at irregular intervals and may depend on clinical decisions, patient condition, workflow and availability of tests or monitoring devices. The way missing values are handled can affect both the statistical properties of the dataset and the performance of machine-learning models.

Recent research has increasingly examined how the missingness mechanism, missingness level and imputation strategy influence downstream clinical prediction. This project will study that relationship in the context of ICU mortality prediction.

2. Problem Statement

To investigate the effect of different missing-data mechanisms and missingness levels on ICU mortality prediction, and to systematically compare commonly used imputation techniques based on both their ability to reconstruct missing values and their effect on downstream classification performance.

3. Aim

The main aim is to determine how the interaction between missingness mechanism, missingness level, imputation method and classification model affects the reliability of ICU mortality prediction.

4. Objectives

- Study the characteristics and patterns of missing data in the selected ICU dataset.
- Model controlled MCAR, MAR and MNAR missingness scenarios at missingness levels from 10% to 70%, to test the robustness of each method.
- Compare KNN, MICE and MissForest with a simple median-imputation baseline.
- Measure imputation quality using MAE, RMSE, R^2 and adjusted R^2 when the original value is known through controlled masking.
- Train and compare machine-learning classifiers on the imputed datasets.
- Evaluate classifiers using AUROC, AUPRC, F1-score, Precision, Log Loss and Brier Score.
- Determine whether lower reconstruction error also results in better downstream mortality prediction.
- Analyze whether the preferred imputation strategy changes with missingness mechanism, missingness level or classifier.

5. Research Questions

- How does increasing missingness affect ICU mortality-prediction performance?
- How does classifier performance differ under MCAR, MAR and MNAR missingness?

- Which imputation method performs best under different missingness mechanisms and levels?
- Does better imputation accuracy necessarily result in better mortality-prediction performance?
- How do imputation methods interact with different classification models?

6. Dataset and Prediction Task

MIMIC-IV will be used as the primary dataset because it is a widely used critical-care EHR database and is directly relevant to the proposed ICU prediction problem. The primary task will be binary ICU mortality prediction.

eICU-CRD may be considered later for external validation if the primary pipeline is completed and project time permits. It is not mandatory for the initial implementation.

7. Missingness Mechanisms

Mechanism	Definition	Proposed modelling
MCAR	Missing Completely At Random	Randomly mask observed values independently of patient characteristics.
MAR	Missing At Random	Make missingness depend on observed variables such as age or other recorded clinical features.
MNAR	Missing Not At Random	Use a documented sensitivity-analysis rule in which missingness depends on the value being masked or an unobserved factor.

The study will not claim that observational data alone can identify the true mechanism. MCAR, MAR and MNAR will instead be represented through controlled masking/simulation so that the experimental conditions are known and reproducible.

8. Imputation Methods

- Median imputation — simple and transparent baseline.
- KNN imputation — estimates missing values using similar observations.
- MICE (Multiple Imputation by Chained Equations) — iterative model-based imputation.
- MissForest — iterative random-forest-based imputation and an important nonlinear baseline.

A modern deep-learning imputation method may be considered later, but it is intentionally excluded from the initial scope to keep the project focused and reproducible.

9. Classification Models

- Logistic Regression — interpretable baseline.
- XGBoost — primary gradient-boosting model.
- CatBoost — comparative gradient-boosting model.
- LightGBM — comparative gradient-boosting model.

NGBoost may be investigated as an optional extension because the project evaluates probabilistic metrics, but it will not be required for the initial experimental pipeline.

10. Proposed Methodology

Implementation will begin only after the literature review is completed and the novelty of the work is finalised.

i) Select the ICU cohort, prediction window, target variable and relevant clinical features from MIMIC-IV.

ii) Perform data cleaning and preprocessing using a leakage-safe train/validation/test split.

iii) Characterize the naturally occurring missing-data pattern in the selected variables.

iv) Create controlled MCAR, MAR and MNAR missingness scenarios across a 10% to 70% span (10%, 20%, 30%, 40%, 50%, 60% and 70%) to test robustness.

v) Apply median, KNN, MICE and MissForest separately to the incomplete datasets.

vi) Evaluate imputation quality by masking values that were originally observed and comparing imputed values with known originals, using MAE, RMSE, R^2 and adjusted R^2.

vii) Train Logistic Regression, XGBoost, CatBoost and LightGBM on each imputed dataset.

viii) Evaluate mortality prediction using discrimination, classification and probability-quality metrics.

ix) Repeat every experiment with at least three random seeds and report mean performance and variability.

x) Compare results across missingness mechanism, missingness level, imputation method and classifier.

11. Evaluation Metrics

Stage	Metrics	Purpose
Imputation	MAE, RMSE, R^2, Adjusted R^2	Measure reconstruction error against known values, and how much of the variation in the known values the imputation recovers.
Classification	AUROC	Measure discrimination between mortality and survival.
Classification	AUPRC	Evaluate positive-class performance, especially under class imbalance.
Classification	F1-Score, Precision	Measure positive-class prediction quality.
Probability quality	Log Loss, Brier Score	Evaluate quality and calibration of predicted mortality probabilities.

$R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \text{mean}(y))^2$ and adjusted $R^2 = 1 - (1 - R^2)(n - 1) / (n - p - 1)$, where y is the known value, $\hat{y}$ the imputed value, n the number of masked entries and p the number of features the imputer uses. All imputation metrics are computed only on masked entries.

12. Experimental Design

The core experiment will evaluate four factors: missingness mechanism, missingness level, imputation method and classifier. The design will use three mechanisms (MCAR, MAR and MNAR), seven controlled missingness levels from 10% to 70% (in steps of 10%), four imputation methods and four classifiers. Using the full 10% to 70% span tests the robustness of each method as missingness increases. Every combination will be repeated with at least three random seeds. A pilot study will be run first to validate the pipeline.

A key analysis will compare imputation quality with downstream predictive utility. A method may have lower RMSE but poorer AUROC, AUPRC or Brier Score than another method; such differences will be analyzed rather than assuming that the best imputation error automatically gives the best classifier.

13. Expected Contribution

The project aims to provide a controlled and reproducible comparison of missing-data handling strategies for ICU mortality prediction. Instead of asking only which imputation method reconstructs missing values most accurately, the study will examine whether imputation quality translates into useful downstream clinical prediction.

The expected outcome is a set of empirical findings describing how imputation strategies behave under different missingness mechanisms and levels, and whether the preferred method changes according to the classifier and evaluation metric.

14. Research Gap

Recent literature has examined missingness mechanisms, imputation quality and clinical prediction. The proposed work connects these stages in a single controlled experiment: missingness mechanism → imputation → mortality prediction. In particular, it will investigate whether methods that perform well according to reconstruction metrics also preserve the information needed for accurate and well-calibrated mortality prediction.

A literature review will be completed first, and the research gap and novelty will be finalised from it before implementation begins.

15. Scope and Exclusions

- The previously proposed fuzzy decision tree and fuzzy membership-function approach is removed from the revised project.
- Genetic-algorithm-based optimization is removed from the revised project.
- The initial study will focus on classical and widely used imputation methods rather than implementing a large number of deep-learning methods.
- MIMIC-IV will be the primary dataset; eICU is an optional extension.
- The initial classifier set will be Logistic Regression, XGBoost, CatBoost and LightGBM.

16. Key Recent References

- A comparative evaluation of handling missing data points and modalities in electronic health records (2026). *International Journal of Medical Informatics*, 210, 106302. Link

- Evaluation of accuracy and potential biases of imputation methods on biomedical datasets subject to different mechanisms and patterns of missingness (2026). *Computer Methods and Programs in Biomedicine: Update*. Link
- Integrated analysis for electronic health records with structured and sporadic missingness (2025). *Journal of Biomedical Informatics*, 171, 104933. Link

17. Initial Work Plan

i) Complete the literature review and finalise the novelty before implementation.

ii) Finalize MIMIC-IV cohort, target definition and feature set.

iii) Set up the git repository and build a leakage-safe preprocessing and evaluation pipeline.

iv) Implement and validate MCAR masking first.

v) Add MAR and MNAR scenarios using documented simulation rules, across the 10% to 70% span.

vi) Implement median, KNN, MICE and MissForest.

vii) Run Logistic Regression and XGBoost as first classifier baselines.

viii) Add CatBoost and LightGBM after the baseline pipeline is validated.

ix) Run all experiments with at least three random seeds.

x) Produce comparison tables and plots for imputation and classification performance.

18. Reproducibility and Version Control

- Every experiment will be run with at least three random seeds (for example 0, 1 and 2). The seed will control the data split, the masking, the imputers and the classifiers.
- The implementation will be designed so that re-running any experiment gives the same result: fixed seeds, saved configuration files for each run, and pinned library versions.
- A git repository will be maintained throughout, with documentation of each implementation step and a change log recording every change (date, author, what changed and why).
- Patient-level MIMIC-IV data will not be committed to the repository.